

Kodre 20/26

Skiciraj krivuljo za Gaussov integral $\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$.

Gaussova porazd.: $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}} \rightarrow P = \int_{-\sigma}^{\sigma} f(x) dx = \frac{2}{\sqrt{2\pi}\sigma} \int_0^{\sigma} e^{-\frac{x^2}{2\sigma^2}} dx =$

$$\left. \begin{array}{l} \frac{x^2}{2\sigma^2} = t^2 \\ \frac{x}{\sqrt{2}\sigma} = t \\ dt = \frac{dx}{\sqrt{2}\sigma} \end{array} \right\} = \frac{2 \cdot \sqrt{2}\sigma}{\sqrt{2\pi}\sigma} \int_0^{1/\sqrt{2}} e^{-t^2} dt = \text{erf}\left(\frac{1}{\sqrt{2}}\right) = 0,67$$

$$\frac{x}{\sqrt{2}\sigma} = t$$

$$dt = \frac{dx}{\sqrt{2}\sigma}$$

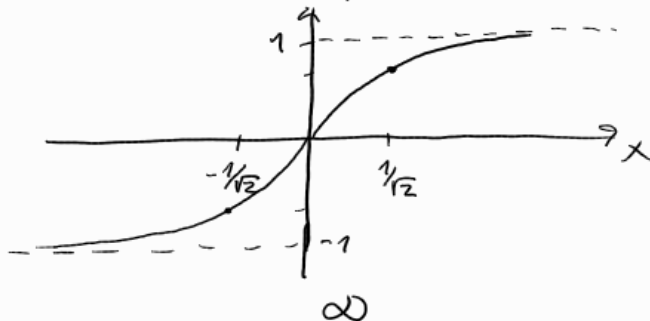
$$\cdot \text{erf}(0) = 0$$

$$\cdot \text{erf}(\infty) = 1$$

• erf monotona naraščajoča, liha

$$\cdot \text{erf}'(x) = \frac{2}{\sqrt{\pi}} e^{-x^2}$$

erf(x)



$$\text{erf}(-x) = \frac{2}{\sqrt{\pi}} \int_0^{-x} e^{-t^2} dt = -\frac{2}{\sqrt{\pi}} \int_0^x e^{-u^2} du = -\text{erf}(x)$$

$t = -u$
 $dt = -du$

• Ali je res $\underbrace{\frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}} dx}_I = 1$?

1. Način: $I^2 = \frac{1}{2\pi\sigma^2} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}} dx \int_{-\infty}^{\infty} e^{-\frac{y^2}{2\sigma^2}} dy =$

pol. koord., $\varphi \in [0, \pi/2]$

$$= \frac{2}{\pi\sigma^2} \int_0^{\infty} \int_0^{\infty} e^{-\left(\frac{x^2+y^2}{2\sigma^2}\right)} dx dy = \frac{2\pi}{\pi\sigma^2} \int_0^{\infty} e^{-\frac{r^2}{2\sigma^2}} r dr =$$

$$= \int_0^{\infty} e^{-u} du = 1 \quad \checkmark$$

$\frac{r^2}{2\sigma^2} = u$

$$\frac{2r dr}{2\sigma^2} = du$$

2. Način

$$I = \dots = \frac{1}{\sqrt{\pi}} \Gamma(1/2) = 1$$

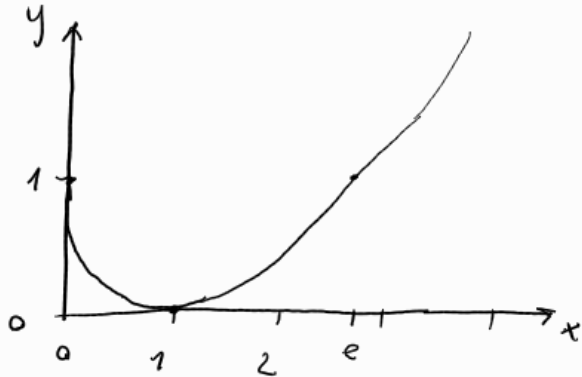
↳ uvedba nove spr.

Kodre 19/8

$x > 0$, skiciraj $y = x \ln x - x + 1$

$$y(0) = 1, y(1) = 0, y(\infty) \rightarrow \infty, y' = \ln x + x \cdot \frac{1}{x} - 1 = \ln x$$

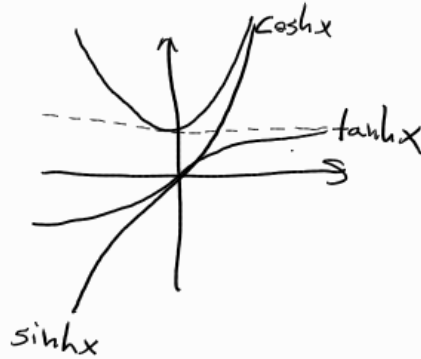
$$y'(0) \rightarrow -\infty, y'(1) = 0, y(e) = 1, y'(e) = 1$$



• zakaj je $y(x)$ zanimiv? $y(x) = \int_1^x \ln t dt$

1.3) Skiciraj $f(x) = \frac{x}{e^{-x}-1} + \frac{x}{2} = \frac{x}{2} \left(1 + \frac{2}{e^{-x}-1} \right) = \frac{x}{2} \frac{2+e^{-x}-1}{e^{-x}-1} = \frac{x(e^{-x}+1)}{2(e^{-x}-1)} =$

$$= -\frac{x}{2} \frac{\cosh x/2}{\sinh x/2}$$

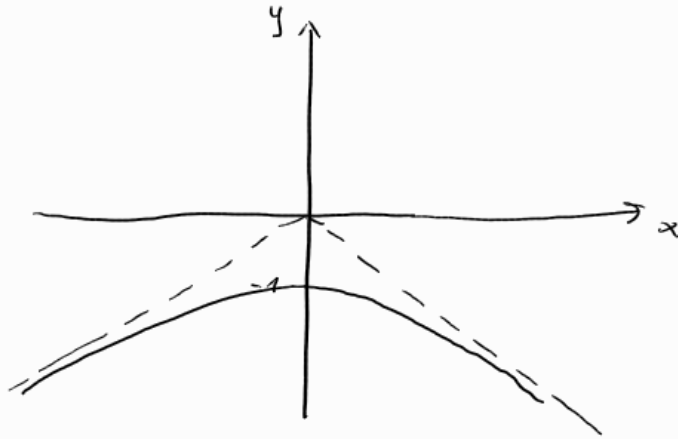


→ soda

$$f(x) = -\frac{x}{2} \frac{1 + \frac{x^2}{8}}{\frac{x}{2} + \frac{x^3}{48}} = -\left(1 + \frac{x^2}{8}\right) \left(1 - \frac{x^2}{24}\right) = \dots = 1 - \frac{x^2}{12} + o(x^3)$$

$$f'(x) = -\frac{x}{6} ; f'(0) = 0$$

$$f(x \rightarrow \infty) \sim -x + \frac{x}{2} = -\frac{x}{2}$$



Kodre 23/6

ura z nihalom, $m=1\text{kg}$, $t_0 \approx 2\text{s}$, $\frac{\delta t_0}{t_0} = 1 \frac{\text{min}}{\text{dan}}$

$\frac{\delta l}{l} = ?$

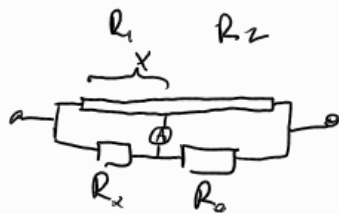
$$t_0 = 2\pi \sqrt{\frac{l}{g}}$$

$$\ln t_0 = \ln 2\pi + \frac{1}{2} \ln l - \frac{1}{2} \ln g \quad / d$$

$$\frac{dt_0}{t_0} = \frac{1}{2} \frac{dl}{l} \rightarrow \frac{dl}{l} = 1,4 \cdot 10^{-3}$$

Kodre 23/11

Wheatstone most



$l=1\text{m}$

$\delta x = 0,3\text{mm}$

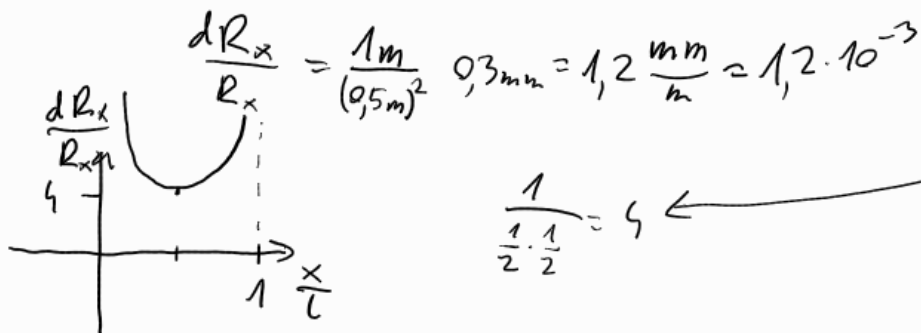
$$\left. \begin{aligned} R_x I_A &= R_1 I_B \\ R_0 I_A &= R_2 I_B \end{aligned} \right\} :$$

$$R_x = \frac{R_1}{R_2} R_0 = \frac{x}{l-x} R_0$$

$$\ln R_x = \ln R_0 + \ln x - \ln(l-x)$$

$$\frac{dR_x}{R_x} = \frac{dx}{x} + \frac{dx}{l-x} = \frac{l}{x(l-x)} dx = \frac{1}{\frac{x}{l}(1-\frac{x}{l})} dx = \frac{dx}{l} \frac{1}{\frac{x}{l}(1-\frac{x}{l})}$$

• Na sredini: $x=l-x=0,5\text{m}$



$$\frac{1}{\frac{1}{2} \cdot \frac{1}{2}} = 4$$

Kodre 23/12

Wheatstoneov most od prej,

$$R_x = R_x(t)$$

$$\alpha = \frac{1}{R} \frac{dR}{dT} = 0,004 \text{ K}^{-1}$$

$$\frac{dx}{dT} = ? \text{ (na sredini)}$$

$$\frac{dR_x}{R_x} = \frac{l}{x(l-x)} dx = \alpha dT$$

$$\frac{dx}{dT} = \frac{\alpha x(l-x)}{l} = \frac{1}{4} \cdot 4 \cdot 10^{-3} \text{ m/K} = 10^{-3} \text{ m/K}$$

Kodre 23/14

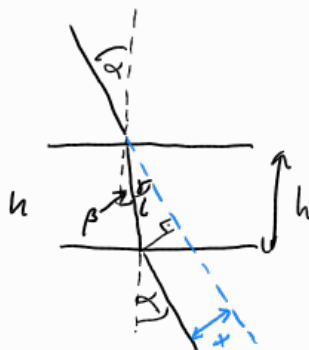
$$d = 5 \text{ cm}$$

$$\alpha = 60^\circ$$

$$\delta\alpha = 0,5^\circ$$

$$n = 1,5$$

$$\delta x = ?$$



$$df(\alpha, \beta) = \frac{\partial f}{\partial \alpha} d\alpha + \frac{\partial f}{\partial \beta} d\beta$$

$$h = l \cos \beta, \quad \gamma = \alpha - \beta$$

$$x = \sin \gamma \cdot l = \frac{h}{\cos \beta} \sin(\alpha - \beta)$$

$$\frac{\sin \alpha}{\sin \beta} = n$$

$$\frac{dx}{h} = d \frac{\sin(\alpha - \beta)}{\cos \beta} = \frac{\cos(\alpha - \beta)}{\cos \beta} d\alpha + \frac{-\cos(\alpha - \beta)(1) \cos \beta + \sin(\alpha - \beta) \sin \beta}{\cos^2 \beta} d\beta$$

$$\left. \begin{array}{l} \sin \alpha = n \sin \beta \\ \cos \alpha d\alpha = n \cos \beta d\beta \\ d\beta = \frac{\cos \alpha}{n \cos \beta} d\alpha \end{array} \right\} \begin{aligned} dx &= h \left[\frac{n \cos^2 \beta \cos(\alpha - \beta) - \cos(\alpha - \beta) \cos \beta \cos \alpha + \sin(\alpha - \beta) \sin \beta \cos \alpha}{n \cos^3 \beta} \right] d\alpha = \\ &= h \left[\frac{\cos(\alpha - \beta) \cos \beta (n \cos \beta - \cos \alpha) + \sin(\alpha - \beta) \sin \beta \cos \alpha}{n \cos^3 \beta} \right] d\alpha \end{aligned}$$

$$\rightarrow dx \rightarrow \delta x, \quad d\alpha \rightarrow \delta\alpha, \quad h = 5 \text{ cm}, \quad n = 1,5; \quad \alpha = 60^\circ; \quad \beta = 35,3^\circ \Rightarrow \text{ustavimo not}$$